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Nose gear spin down apparatus

Abstract

A nose gear spin down apparatus includes a spin down device and a resistance device. The spin down device is rotatable about a central longitudinal axis. The spin down device rotates about the central longitudinal axis after engagement with a spinning nose gear wheel of an aircraft. The resistance device contacts the spin down device without imparting a resistance to the spin down device. The resistance device imparts the resistance to the spin down device after conditions are met.

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Background/Summary

BACKGROUND INFORMATION

1. Field

(1) The present disclosure relates generally to nose gear wheel stowage. More specifically, the present disclosure relates to a nose gear spin down apparatus for arresting rotation of a nose gear wheel when it is stowed after takeoff.

2. Background

(2) The nose gear wheel of an aircraft after takeoff spins at a significant rate (takeoff speed). A typical solution for stopping rotation of a nose gear wheel of an aircraft after takeoff employs a non-moving brake or metallic cleat mounted in the nose gear bay. After takeoff, the nose gear is retracted into the nose gear bay or wheel well such that the nose gear wheel comes into contact with a cleat. The cleat is stationary and the cleat rubs against the nose gear wheel tire until the wheel's rotation is arrested by friction. The nose gear wheel is arrested within seconds of making contact with the cleat.

(3) A major drawback to the existing solution is that the contact/rubbing between the tire and cleat creates a significant amount of tire debris that accumulates on all surfaces inside the nose gear bay. The accumulation of rubber particles within the nose gear bay reduces the ability to inspect or visibility of structural parts housed within the nose gear bay. Lack of ability to inspect hinders airplane maintenance and can mask the existence of cracks or other damage. The tire debris build up thus requires consistent removal and regular cleaning of the nose gear bay. Also, the tire undergoes additional wear as the rubber of the tire is eroded with each takeoff ultimately reducing the lifecycle of the nose gear tire.

(4) Therefore, it would be desirable to have a method and apparatus that takes into account at least some of the issues discussed above, as well as other possible issues.

SUMMARY

(5) An illustrative embodiment of the present disclosure provides a nose gear spin down apparatus comprising a spin down device and a resistance device. The spin down device is rotatable about a central longitudinal axis. The spin down device rotates about the central longitudinal axis at a first speed after engagement with a spinning nose gear wheel of an aircraft. The spinning nose gear wheel of the aircraft spins at a second speed. The resistance device contacts the spin down device without imparting a resistance to the spin down device. The resistance device imparts the resistance to the spin down device when the first speed of the spin down device matches the second speed of the spinning nose gear wheel of the aircraft.

(6) Another illustrative embodiment of the present disclosure provides nose gear spin down apparatus comprising a tubular casing, a threaded shaft, and a resistance device. The tubular casing includes internal threads. The threaded shaft is positioned within the tubular casing. External threads of the threaded shaft engage the internal threads of the tubular casing. The resistance device is axially aligned with the threaded shaft and the tubular casing, the threaded shaft, and the resistance device all share a central longitudinal axis. Movement of the threaded shaft relative to the tubular casing along the central longitudinal axis or movement of the tubular casing relative to the threaded shaft along the central longitudinal axis engages the resistance device after the threaded shaft or the tubular casing moves through a space.

(7) A further illustrative embodiment of the present disclosure provides a method for slowing rotation of a nose gear wheel of an aircraft after takeoff. The method includes a step of retracting a spinning nose gear wheel of the aircraft to a stowed position in a nose gear bay of the aircraft after takeoff of the aircraft. The method includes another step of engaging the spinning nose gear wheel of the aircraft with a nose gear spin down apparatus mounted in the nose gear bay of the aircraft without imparting a resistance to the spinning nose gear wheel of the aircraft. The method further includes a step of imparting the resistance to the spinning nose gear wheel of the aircraft with a resistance device of the nose gear spin down apparatus after an element of the nose gear spin down apparatus moves through a space between the element of the nose gear spin down apparatus and the resistance device and into contact with the resistance device.

(8) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is an illustration of an aircraft in accordance with an illustrative example;

(3) FIG. 2 is an illustration of a block diagram of an nose gear spin down apparatus and an aircraft in accordance with an illustrative embodiment;

(4) FIG. 3 is an illustration of nose gear spin down apparatus in accordance with an illustrative embodiment;

(5) FIG. 4 is an illustration of a cross-section view of a nose gear spin down apparatus in accordance with an illustrative embodiment;

- (6) FIG. 5 is an illustration of a nose gear spin down apparatus applying a resistance in accordance with an illustrative embodiment;
- (7) FIG. 6 is an illustration of a nose gear spin down apparatus in accordance with an illustrative embodiment;
- (8) FIG. 7 is an illustration of a nose gear spin down apparatus in accordance with an illustrative embodiment;
- (9) FIG. 8 is an illustration of a nose gear spin down apparatus in accordance with an illustrative embodiment;
- (10) FIG. 9 is an illustration of a nose gear spin down apparatus in accordance with an illustrative embodiment;
- (11) FIG. 10 is an illustration of a side view of a nose gear spin down apparatus in accordance with an illustrative embodiment;
- (12) FIG. 11 is an illustration of a nose gear spin down apparatus applying a resistance in accordance with an illustrative embodiment;
- (13) FIG. 12 is an illustration of a flowchart of a process for slowing rotation of a nose gear wheel of an aircraft after takeoff in accordance with an illustrative embodiment;
- (14) FIG. 13 is an illustration of a block diagram of an aircraft manufacturing and service method in accordance with an illustrative embodiment; and
- (15) FIG. 14 is an illustration of a block diagram of an aircraft in which an illustrative embodiment may be implemented.

DETAILED DESCRIPTION

- (16) The illustrative embodiments recognize and take into account one or more different considerations. For example, the illustrative embodiments recognize and take into account that nose gear wheels of aircraft spin at a high rate of speed just after takeoff and before being retracted into a nose gear bay.
- (17) The illustrative embodiments recognize and take into account that there is a need to arrest the spinning nose gear wheel from spinning after takeoff and after stowage.
- (18) The illustrative embodiments recognize and take into account that current mechanisms installed on today's aircraft include a stationary cleat that contacts the spinning nose gear wheel once the nose gear wheel is retracted to a stowed position within the nose gear bay.
- (19) The illustrative embodiments recognize and take into account that contact between the spinning nose gear wheel and a stationary cleat mounted inside the nose gear bay causes undue wear on the nose gear wheel itself and causes debris accumulation in the nose gear bay inhibiting inspection and maintenance.
- (20) Thus, the illustrative embodiments provide a nose gear spin down apparatus that provides a resistance to the spinning nose gear wheel to arrest the spinning nose gear wheel without putting undue wear and tear on the nose gear wheel.
- (21) With reference now to the figures, and in particular, with reference to FIG. 1, an illustration of an aircraft is depicted in accordance with an illustrative example. In this illustrative example, aircraft **100** has wing **102** and wing **104** attached to fuselage **106**. Aircraft **100** includes engine **108** attached to wing **102** and engine **110** attached to wing **104**.
- (22) Fuselage **106** has tail section **112**. Horizontal stabilizer **114**, horizontal stabilizer **116**, and vertical stabilizer **118** are attached to tail section **112** of fuselage **106**.
- (23) Aircraft **100** is an example of an aircraft in which a nose gear spin down apparatus may be installed in accordance with an illustrative example. For example, a nose gear bay located in fuselage **106** of aircraft **100** may have a nose gear spin down apparatus mounted therein. Nose gear wheels after takeoff spin at a high rate of speed. It is necessary to stop the spinning of the nose gear wheels after they have been retracted into the nose gear bay of the aircraft. A nose gear spin down apparatus as disclosed provides a resistance to the spinning nose gear wheel in order to dissipate rotation of the nose gear wheel without providing undue wear and tear on the nose gear wheel

while also alleviating the need for time consuming and costly maintenance/cleaning of the nose gear bay.

(24) The illustration of aircraft **100** in FIG. **1** is not meant to imply physical or architectural limitations to the manner in which an illustrative example may be implemented. For example, although aircraft **100** is a commercial aircraft, aircraft **100** may be a military aircraft, a rotorcraft, a helicopter, an unmanned aerial vehicle, or any other suitable aircraft.

(25) Although the illustrative examples for an illustrative example are described with respect to an aircraft, the illustrative example may be applied to other types of platforms. The platform may be, for example, a mobile platform, a stationary platform, a land-based structure, an aquatic-based structure, or a space-based structure. More specifically, the platform may be an aircraft, a surface ship, a tank, a personnel carrier, a train, a spacecraft, a space station, a satellite, a submarine, an automobile, a power plant, a bridge, a dam, a house, a manufacturing facility, a building, a tool, a mechanical structure, or some other suitable platform or structure where installation of nose gear spin down apparatus is desirable.

(26) Turning now to FIG. **2**, an illustration of a block diagram of aircraft **202** and nose gear spin down apparatus **204** is depicted in accordance with an illustrative example.

(27) Aircraft **202** includes spinning nose gear wheel **206** which is retractable and deployable within and from nose gear bay **208**. Spinning nose gear wheel **206** spins in forward direction **210** at speed **212**. Speed **212** is generally based on or is a function of take off speed of aircraft **202**, depending on tire diameter. Once aircraft **202** takes off, spinning nose gear wheel **206** spins at speed **212** in forward direction **210** until the rotation is arrested. Of course, nose gear wheel **206**, unless acted on by a cleat or the disclosed nose gear spin down apparatus **204**, would eventually stop spinning on its own due to environmental forces that act upon it (e.g., gravity, friction, air resistance). However, it is desirable to arrest the rotation of spinning nose gear wheel **206** promptly to prevent any unnecessary centrifugal forces from spinning nose gear wheel **206** acting on aircraft **202** and to prevent possible damage to the aircraft systems located in the nose gear bay from rotating tire debris should something happen to the tire. If a tire is damaged, then centrifugal forces from a spinning nose gear wheel could eject tire debris.

(28) Spinning nose gear wheel **206** is retractable within and deployable from nose gear bay **208** between stowed position **214** and deployed position **216**. When in stowed position **214**, spinning nose gear wheel **206** contacts nose gear spin down apparatus **204** mounted in nose gear bay **208**. Nose gear spin down apparatus **204** dissipates the rotation energy of spinning nose gear wheel **206** from speed **212** to a stationary state. When in deployed position **216**, spinning nose gear wheel **206** is free to rotate without any interference from nose gear spin down apparatus **204**.

(29) Nose gear spin down apparatus **204** has spin down device **220** and resistance device **222** in this illustrative example. Spin down device **220** may be spaced from resistance device **222** by space **224**.

(30) Spin down device **220** contacts spinning nose gear wheel **206** when spinning nose gear wheel **206** is in stowed position **214** within nose gear bay **208**. Of course, it is understood that spinning nose gear wheel **206** will begin slowing and eventually cease spinning once spinning nose gear wheel **206** comes into contact with spin down device **220**.

(31) In this illustrative example, spin down device **220** includes tubular casing **226**. Tubular casing **226** rotates about central longitudinal axis **228**. Tubular casing **226** is urged to rotate about central longitudinal axis **228** from contact with spinning nose gear wheel **206**. In other words, the rotation of spinning nose gear wheel **206** causes rotation of tubular casing **226** once spinning nose gear wheel **206** contacts tubular casing **226**. Once spinning nose gear wheel **206** contacts tubular casing **226**, tubular casing **226** rotates about central longitudinal axis **228** at speed **230**. Tubular casing **226** may include high friction outer surface **232** to help gain traction for contact with spinning nose gear wheel **206**. High friction outer surface **232** may be a machined surface or an adhesive surface, for example.

(32) Tubular casing **226** may include wheel **234**. Wheel **234** may be mounted on tubular casing **226**. When wheel **234** is mounted on tubular casing **226**, wheel **234** rotates about central longitudinal axis **228**. When wheel **234** is mounted on tubular casing **226**, wheel **234** contacts spinning nose gear wheel **206** when spinning nose gear wheel **206** is retracted into stowed position **214**. When wheel **234** contacts spinning nose gear wheel **206**, tubular casing does not necessarily contact spinning nose gear wheel **206**. When wheel **234** contacts spinning nose gear wheel **206**, wheel **234** rotates about central longitudinal axis **228** at speed **230**.

(33) Tubular casing **226** is hollow and further includes internal threads **236**.

(34) Spin down device **220** includes threaded shaft **240**. Threaded shaft **240** also has central longitudinal axis **228**. Threaded shaft **240** is housed within tubular casing **226**. Tubular casing **226** and threaded shaft **240** share central longitudinal axis **228**. Threaded shaft **240** includes external threads **242**. External threads **242** engage internal threads **236** of tubular casing **226**. Threaded shaft **240** may also include gear set **244**.

(35) Spin down device **220** may also include axle **246**. When spin down device **220** includes axle **246**, wheel **234** is mounted on axle **246**. When spin down device **220** includes axle **246** and wheel **234**, axle **246** and wheel **234** rotate at speed **230** when wheel **234** contacts spinning nose gear wheel **206**.

(36) Axle **246** includes gear set **248**. Gear set **248** includes forward gear **250** and reverse gear **252** mounted on axle **246**. Gear set **248** is slidable **254** along axle **246**. Gear set **248** is slidable **254** along axle **246** between forward position **256** and reverse position **258**.

(37) In this illustrative example, resistance device **222** provides resistance **260** to spin down device **220**. Resistance **260** ultimately arrests rotation of spinning nose gear wheel **206** when spinning nose gear wheel **206** is in stowed position **214** in nose gear bay **208**. Resistance device **222** provides resistance to an element of spin down device **220** that contacts resistance device **222**. For example, upon rotation of tubular casing **226** resulting from contact with spinning nose gear wheel **206**, the rotation of tubular casing **226** and the engagement of internal threads **236** with external threads **242** of threaded shaft **240** urges movement of threaded shaft **240** relative to tubular casing **226** such that threaded shaft **240** moves parallel to central longitudinal axis **228** through space **224** and into contact with resistance device **222**. Resistance device **222** provides resistance **260** to retard the longitudinal movement of threaded shaft **240**. As the longitudinal movement of threaded shaft **240** slows, the engagement of external threads **242** with internal threads **236** of tubular casing **226** slows speed **230** of tubular casing **226**. Because tubular casing **226**, or wheel **234** mounted to tubular casing **226** of spin down device **220**, is in contact with spinning nose gear wheel **206**, the slowing of speed **230** of spin down device **220** slows speed **212** of spinning nose gear wheel **206** until spinning nose gear wheel **206** stops spinning.

(38) It is important to note that resistance device **222** imparts no resistance to the spin down device at initial contact between spin down device **220** and spinning nose gear wheel **206**. The free unhindered rotation of the spin down device at first contact is to avoid the undue wear on the rubber tire of a spinning nose gear wheel that occurs when spinning nose gear wheel contacts a static cleat. Resistance device **222** imparts no resistance to the spin down device until an element of the spin down device moves through space **224** and contacts resistance device **222** or as speed **230** of spin down device **220** matches speed **212** of spinning nose gear wheel **206**.

(39) Resistance device **222** may provide resistance **260** to spin down device **220** via spring **262**. Resistance device **222** may provide resistance **260** to spin down device **220** via series of springs **264**. Each spring of series of springs **264** may have a different stiffness **266**. Spring **262** or series of spring **264** can provide progressive resistance **268** to spin down device **220**. Resistance device **222** may provide resistance **260** to spin down device **220** via viscous damper **270**. Non-limiting examples of viscous damper **270** include rotary damper **272** and spring loaded viscous damper **274**.

(40) When incorporating spring **262** or series of springs **264**, resistance device **222** may include locking device **276**. As resistance device **222** provides resistance **260** to spin down device **220**,

spring **262** or series of springs **264** compress. Once rotation of spinning nose gear wheel **206** is stopped, locking device **276** may engage resistance device **222** to maintain spring **262** or series of springs **264** compressed to capture stored energy **278**. In an illustrated example, stored energy **278** may be used to spin up a previously stopped nose gear wheel in forward direction **210** before the nose gear wheel is deployed to deployed position **216** in preparation for landing. Spinning the nose gear wheel back up to near takeoff speed will assist in reducing wear and tire on the rubber tire of the nose gear wheel.

(41) As used herein, a first component “connected to” or “coupled to” or “associated with” a second component means that the first component can be connected directly or indirectly to the second component. The connection is a physical association. In other words, additional components may be present between the first component and the second component. The first component is considered to be indirectly connected to the second component when one or more additional components are present between the two components. When the first component is directly connected to the second component, no additional components are present between the two components.

(42) For example, a first component can be considered to be physically connected to a second component by at least one of being secured to the second component, bonded to the second component, mounted to the second component, welded to the second component, fastened to the second component, or connected to the second component in some other suitable manner. The first component also can be connected to the second component using a third component. The first component can also be considered to be physically connected to the second component by being formed as part of the second component, an extension of the second component, or both.

(43) As used herein, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

(44) For example, without limitation, “at least one of item A, item B, or item C” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

(45) With reference next to FIGS. **3-5**, illustrations of a nose gear spin down apparatus are depicted in accordance with an illustrative example. In this illustrative example, the components illustrated in FIGS. **3-5** are examples of nose gear spin down apparatus **204** and spinning nose gear wheel **206** shown in block form in FIG. **2**.

(46) As depicted, nose gear spin down apparatus **300** has yet to come in contact with spinning nose gear wheel **301**. Nose gear spin down apparatus **300** includes spin down device **302** and resistance device **304**.

(47) In this illustrative example, spin down device **302** includes tubular casing **306**. Tubular casing **306** rotates about central longitudinal axis **308**. Tubular casing **306** is hollow and includes internal threads **310**. Spin down device **302** includes wheel **312**. Wheel **312** is mounted on tubular casing **306**. In other words, as wheel **312** rotates, tubular casing **306** rotates at the same rotation speed. Wheel **312** rotates about central longitudinal axis **308**. Wheel **312** and tubular casing **306** are urged to rotate about central longitudinal axis **308** from contact with a spinning nose gear wheel at a speed that eventually matches the speed of the spinning nose gear wheel. Wheel **312** contacts a spinning nose gear wheel when a spinning nose gear wheel is retracted into a stowed position within a nose gear bay of an aircraft.

(48) In an alternate example, wheel **312** may not be present. Tubular casing **306** is urged to rotate about central longitudinal axis **308** from direct contact with a spinning nose gear wheel in lieu of

wheel **312**. High friction surface **314** of tubular casing **306** in contact with a spinning nose gear wheel causes rotation of tubular casing **306**. Once a spinning nose gear wheel contacts tubular casing **306**, tubular casing **306** rotates about central longitudinal axis **308** at speed that eventually matches the speed of the spinning nose gear wheel. High friction surface **314** may be a machined surface or an adhesive surface, for example.

(49) Spin down device **302** includes threaded shaft **316**. Threaded shaft **316** also has central longitudinal axis **308**. Threaded shaft **316** includes external threads **318**. External threads **318** are sized to engage internal threads **310** of tubular casing **306**.

(50) In this illustrative example, resistance device **304** is spring **320**. Spring **320** is spaced from threaded shaft **316**, or spacer **326** if present, by space **328**. It is important to note that prior to spin down device **302** contacting a spinning nose gear wheel, spring **320** is spaced from threaded shaft **316** whether spacer **326** is present or not. In other words, spring **320** does not contact threaded shaft **316** before spin down device contacts a spinning nose gear wheel of an aircraft. When spring **320** is in a resting, non-compressed state, spring **320** is spaced from threaded shaft **316** by at least space **328**.

(51) Brackets **322**, **323** are used to mount nose gear spin down apparatus **300** to the inside of a nose gear bay. Center rod **324** is connected to and extends between brackets **322**, **323**. Threaded shaft **316** is slidable in direction **330** along center rod **324**. Spring **320** is compressible and expandable in direction **330** along center rod **324**. Center rod **324** has non-circular cross-section **402**. Non-circular cross-section **402** prevents threaded shaft **316** from rotating about central longitudinal axis **308**. Spring **320** is coiled around center rod **324**.

(52) In use, nose gear spin down apparatus **300** arrests the rotation of a spinning nose gear wheel of an aircraft when the spinning nose gear wheel is retracted to a stowed position in a nose gear bay of the aircraft after takeoff of the aircraft.

(53) Once retracted into the nose gear bay of the aircraft, the spinning nose gear wheel contacts spin down device **302**. The spinning nose gear wheel contacts wheel **312** or contacts tubular casing **306** if wheel **312** is not present. Rotation of the spinning nose gear wheel causes wheel **312** and tubular casing **306** or just tubular casing **306** (if wheel **312** is not present) to rotate about central longitudinal axis **308** in an opposite direction than the rotation of the spinning nose gear wheel. The direction of the spinning nose gear wheel can be considered a forward direction. Rotation of tubular casing **306** and the engagement of internal threads **310** with external threads **318** of threaded shaft **316** causes threaded shaft **316** to translate longitudinally along center rod **324** in direction **502**. Translation of threaded shaft **316** in direction **502** causes threaded shaft **316** to move through space **328** and into contact with spring **320**. As spring **320** compresses due to the movement of threaded shaft **316**, spring **320** provides resistance **504** to slow down the movement of threaded shaft in direction **502**. Resistance **504** may be progressive based on the distance threaded shaft translates. Slowing the movement of the threaded shaft in direction **502** causes the rotation speed of tubular casing **306** to slow which ultimately slows and finally stops the spinning nose gear wheel that is in contact with nose gear spin down apparatus **300**.

(54) It is important to note that resistance device **304** imparts no resistance to spin down device **302** at initial contact between spin down device **302** and a spinning nose gear wheel. Spring **320** provides no resistance **504** until threaded shaft **316** (or spacer **326**, if present) contacts spring **320**. An element of spin down device **302**, whether that be threaded shaft **316** or spacer **326**, must move through space **328** and into contact with spring **320** before spring **320** provides resistance **504**. Because spring **320** does not provide resistance immediately, initial contact of the spinning nose gear wheel with the spin down device does not rub the tire of the nose gear wheel and does not cause undue wear on the tire of the nose gear wheel. The speed of the spinning nose gear wheel is matched by the speed of the spin down device before resistance is imparted by the spring on the spin down device.

(55) With reference next to FIG. 6, illustration of a nose gear spin down apparatus is depicted in

accordance with an illustrative example. In this illustrative example, nose gear spin down apparatus **600** is an example of one implementation for nose gear spin down apparatus **204** shown in block form in FIG. 2.

(56) As depicted, nose gear spin down apparatus **600** includes spin down device **602** and resistance device **604**.

(57) In this illustrative example, spin down device **602** includes tubular casing **606**. Tubular casing **606** rotates about central longitudinal axis **608**. Tubular casing **606** is hollow and includes internal threads **610**. Spin down device **602** includes wheel **612**. Wheel **612** is connected to tubular casing **606**. As wheel **612** rotates, tubular casing **606** rotates at the same rotation speed. Wheel **612** rotates about central longitudinal axis **608**. Wheel **612** and tubular casing **606** are urged to rotate about central longitudinal axis **608** from contact with a spinning nose gear wheel at a speed that eventually matches the speed of the spinning nose gear wheel. Wheel **612** contacts a spinning nose gear wheel when a spinning nose gear wheel is retracted into a stowed position within a nose gear bay of an aircraft.

(58) In this illustrative example, resistance device **604** is viscous damper **614**. In non-limiting examples, viscous damper **614** may be a rotary damper or a spring-loaded viscous damper. When viscous damper **614** is in a resting state, viscous damper **614** provides no resistance to spin down device **602**. Threaded shaft **616** is connected to piston rod **618**. Piston head **620** is mounted to piston rod **618**.

(59) As described previously, rotation of tubular casing **606** causes threaded shaft **616** to translate along central longitudinal axis **608** in direction **622**. Threaded shaft **616** moving in direction **622** pushes piston head **620** through the viscous fluid contained within viscous damper **614**. Viscous damper **614** provides no resistance **624** to spin down device **602** while piston rod **618** and piston head **620** are at rest within viscous damper **614**. After piston head **620** reaches a certain velocity within viscous damper **614**, then viscous damper **614** provides resistance **624**. Slowing the movement of the threaded shaft in direction **622** causes the rotation speed of tubular casing **606** to slow which ultimately slows and finally stops the spinning nose gear wheel that is in contact with nose gear spin down apparatus **600**. In this illustrative example, viscous damper **614** may be a sealed cylinder that returns to a neutral position when unloaded such as a spring loaded viscous damper. In this illustrative example, viscous damper **614** may be a rotary damper that rotates when acted on by threaded shaft **616**.

(60) Alternatively, threaded shaft **616** may not be connected to piston rod **618**. Thus, threaded shaft **616** is distanced from piston rod by a space. Translation of threaded shaft **616** in direction **622** causes threaded shaft **616** to move through the space between threaded shaft **616** and piston rod **618** and into contact with piston rod **618**. Once threaded shaft **616** contacts piston rod **618**, viscous damper **614** provides resistance **624** to slow down the movement of threaded shaft **616** in direction **622**. Slowing the movement of the threaded shaft in direction **622** causes the speed of rotation of tubular casing **606** to slow which ultimately slows and finally stops the spinning nose gear wheel that is in contact with nose gear spin down apparatus **600**.

(61) It is important to note that resistance device **604** imparts no resistance to spin down device **602** at initial contact between spin down device **602** and a spinning nose gear wheel. Viscous damper **614** provides no resistance **624** until piston rod **618** reaches a certain velocity or, alternatively, threaded shaft **616** contacts piston rod **618**.

(62) With reference next to FIG. 7, illustration of a nose gear spin down apparatus is depicted in accordance with an illustrative example. In this illustrative example, nose gear spin down apparatus **700** is an example of one implementation for nose gear spin down apparatus **204** shown in block form in FIG. 2.

(63) As depicted, nose gear spin down apparatus **700** includes spin down device **702** and resistance device **704**.

(64) In this illustrative example, spin down device **702** includes wheel **706** mounted to axle **708**.

Axle **708** and wheel **706** rotate about central longitudinal axis **710**. Resistance device **704** includes brake disc **712** and brake pad **714**. Wheel **706** contacts a spinning nose gear wheel when a spinning nose gear wheel is retracted into a stowed position within a nose gear bay of an aircraft. Brake pad **714** pressed against brake disc **712** provides resistance to the rotation of wheel **706** through friction. The friction provided can be a constant friction or a progressive friction based on speed of rotation of the spinning nose gear wheel. Slowing the rotation of wheel **706** ultimately arrests rotation of the spinning nose gear wheel that spin down device is in contact with.

(65) With reference next to FIGS. **8-11**, illustrations of a nose gear spin down apparatus are depicted in accordance with an illustrative example. In this illustrative example, the components illustrated in FIGS. **8-11** are examples of nose gear spin down apparatus **204** and spinning nose gear wheel **206** shown in block form in FIG. **2**.

(66) As depicted, nose gear spin down apparatus **800** has yet to come in contact with spinning nose gear wheel **801**. Nose gear spin down apparatus **800** includes spin down device **802** and resistance device **804**.

(67) Spin down device **802** includes wheel **806** mounted to axle **808**. Wheel **806** and axle **808** rotate about central longitudinal axis **810**. Wheel **806** and axle **808** are urged to rotate about central longitudinal axis **810** from contact with a spinning nose gear wheel at a speed that eventually matches the speed of the spinning nose gear wheel. Wheel **806** contacts a spinning nose gear wheel when a spinning nose gear wheel is retracted into a stowed position within a nose gear bay of an aircraft.

(68) Axle **808** includes gear set **812**. Gear set **812** includes forward gear **814** and reverse gear **816** mounted on axle **808**. Gear set **812** is slidable in direction **820** along axle **808**. Gear set **812** is slidable in direction **820** along axle **808** between forward position **822** (as depicted in FIG. **8**) and reverse position **824** (as depicted in FIG. **9**). Gear shift arm **826** acts on gear set **812** to move gear set **812** between forward position **822** and reverse position **824**.

(69) Spin down device **802** includes threaded shaft **830**. Threaded shaft **830** includes gear set **832**. Threaded shaft **830** is housed within tubular casing **834**. Threaded shaft has external threads **836**. External threads **836** are sized to engage internal threads **840** of tubular casing **834**. Threaded shaft **830** rotates about central longitudinal axis **842**.

(70) Resistance device **804** includes spring **850** housed within casing **852**. Casing **852** is sized to accept tubular casing **834**. Spring **850** provides resistance **856** in direction **858**.

(71) Spin down device **802** further includes idler gear **860**. When gear set **812** is in forward position **822**, forward gear **814** is engaged with gear set **832** of threaded shaft **830**. When gear set **812** is in reverse position **824**, reverse gear **816** is engaged with idler gear **860** and idler gear **860** is engaged with gear set **832** of threaded shaft **830**.

(72) In use, nose gear spin down apparatus **800** arrests the rotation of a spinning nose gear wheel of an aircraft when the spinning nose gear wheel is retracted to a stowed position in a nose gear bay of the aircraft after takeoff of the aircraft.

(73) Once retracted into the nose gear bay of the aircraft, the spinning nose gear wheel contacts spin down device **802**. Specifically, the spinning nose gear wheel contacts wheel **806** of spin down device **802**. Rotation of the spinning nose gear wheel causes wheel **806** and axle **808** to rotate about central longitudinal axis **810** in an opposite direction than the rotation of the spinning nose gear wheel. The direction of the spinning nose gear wheel can be considered a forward direction.

(74) When gear set **812** is in forward position **822**, gear set **812** engaged with gear set **832** of threaded shaft **830** rotates threaded shaft **830** about central longitudinal axis **842** in the same forward direction as the spinning nose gear wheel. Rotation of threaded shaft **830** and the engagement of external threads **836** with internal threads **840** of tubular casing **834** causes tubular casing **834** to move along central longitudinal axis **842** in direction **862**. Movement of tubular casing **834** in direction **862** causes tubular casing **834** to contact spring **850**. As spring **850** compresses due to the movement of tubular casing **834**, spring **850** provides resistance **856** in

direction **858** to slow down the movement of tubular casing **834** in direction **862**. Slowing the movement of the tubular casing in direction **862** causes the rotation speed of threaded shaft **830** to slow which ultimately slows and finally stops the spinning nose gear wheel that is in contact with nose gear spin down apparatus **800**.

(75) With the rotation of the spinning nose gear stopped, a locking mechanism of resistance device **804** retains spring **850** in a compressed state thus storing energy in spring **850** received from the spinning nose gear wheel. Prior to deploying the nose gear wheel from the stowed position to a deployed position in preparation for landing, nose gear spin down apparatus **800** can release the stored energy and spin up the nose gear wheel in a forward direction as the nose gear wheel is deployed. Spinning up the nose gear wheel in the forward direction is beneficial as it further reduces wear on the rubber tire of the nose gear wheel by reducing friction between the nose gear wheel and the landing surface.

(76) To spin up the nose gear wheel, gear shift arm **826** acts on gear set **812** to move gear set **812** from forward position **822** to reverse position **824**. When gear set **812** is in reverse position **824**, reverse gear **816** is engaged with idler gear **860** and idler gear **860** is engaged with gear set **832** of threaded shaft **830**. The locking mechanism releases spring **850** such that spring **850** urges tubular casing **834** in direction **858** which rotates threaded shaft **830**, which rotates axle **808** and wheel **806**, which ultimately rotates the nose gear wheel in the forward direction.

(77) With reference next to FIG. **12**, an illustration of a flowchart of a process **1200** for slowing rotation of a nose gear wheel of an aircraft after takeoff is depicted in accordance with an illustrative example. The method depicted in FIG. **12** may be used in conjunction with nose gear spin down apparatus depicted in FIGS. **1-11**.

(78) The process begins by retracting a spinning nose gear wheel of an aircraft to a stowed position in a nose gear bay of the aircraft after takeoff of the aircraft (operation **1202**). The process engages the spinning nose gear wheel of the aircraft with a nose gear spin down apparatus. The nose gear spin down apparatus is mounted in the nose gear bay of the aircraft. Initially, the nose gear spin down apparatus does not impart a resistance to the spinning nose gear wheel (operation **1204**). The process then imparts the resistance to the spinning nose gear wheel of the aircraft with a resistance device of the nose gear spin down apparatus after an element of the nose gear spin down apparatus moves through a space between the element of the nose gear spin down apparatus and the resistance device and into contact with the resistance device (operation **1206**). The process stores energy from the spinning nose gear wheel as the nose gear spin down apparatus imparts the resistance to the spinning nose gear wheel and slows rotation of the nose gear wheel after takeoff (operation **1208**). Prior to deploying the nose gear wheel of the aircraft from the stowed position or during deploying the nose gear wheel of the aircraft from the stowed position, the process releases the stored energy to the nose gear wheel to rotate the nose gear wheel in the forward direction in preparation for landing (operation **1210**). The nose gear spin down apparatus can release the stored energy to the nose gear wheel up until the point that the nose gear wheel and the nose gear spin down apparatus disengage.

(79) In some alternative implementations of an illustrative example, the function or functions noted in the blocks may occur out of the order noted in the Figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram.

(80) The illustrative embodiments of the disclosure may be further described in the context of aircraft manufacturing and service method **1300** as shown in FIG. **13** and aircraft **1400** as shown in FIG. **14**. Turning first to FIG. **13**, an illustration of a block diagram of an aircraft manufacturing and service method is depicted in accordance with an illustrative embodiment. During pre-production, aircraft manufacturing and service method **1300** may include specification and design **1302** of aircraft **1400** in FIG. **14** and material procurement **1304**.

(81) During production, component and subassembly manufacturing **1306** and system integration **1308** of aircraft **1400** in FIG. **14** takes place. Thereafter, aircraft **1400** in FIG. **14** may go through certification and delivery **1310** in order to be placed in service **1312**. While in service **1312** by a customer, aircraft **1400** in FIG. **14** is scheduled for routine maintenance and service **1314**, which may include modification, reconfiguration, refurbishment, and other maintenance, service, or inspection.

(82) The apparatus of this disclosure may be installed on an aircraft during component and subassembly manufacturing **1306**. In addition, the apparatus of this disclosure may be retrofitted onto aircraft **1400** in FIG. **14** during routine maintenance and service **1314** as part of a modification, reconfiguration, or refurbishment of aircraft **1400** in FIG. **14**.

(83) Each of the processes of aircraft manufacturing and service method **1300** may be performed or carried out by a system integrator, a third party, an operator, or some combination thereof. In these examples, the operator may be a customer. For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors, and suppliers, and an operator may be an airline, a leasing company, a military entity, a service organization, and so on.

(84) With reference now to FIG. **14**, an illustration of a block diagram of an aircraft is depicted in which an illustrative embodiment may be implemented. In this example, aircraft **1400** is produced by aircraft manufacturing and service method **1300** in FIG. **13** and may include airframe **1402** with plurality of systems **1404** and interior **1406**. Examples of systems **1404** include one or more of propulsion system **1408**, electrical system **1410**, hydraulic system **1412**, and environmental system **1414**. Any number of other systems may be included. Although an aerospace example is shown, different illustrative embodiments may be applied to other industries, such as the automotive industry.

(85) Apparatuses and methods embodied herein may be employed during at least one of the stages of aircraft manufacturing and service method **1300** in FIG. **13**. In one illustrative example, components or subassemblies produced in component and subassembly manufacturing **1306** in FIG. **13** may be fabricated or manufactured in a manner similar to components or subassemblies produced while aircraft **1400** is in service **1312** in FIG. **13**. As yet another example, one or more apparatus embodiments, method embodiments, or a combination thereof may be utilized during production stages, such as component and subassembly manufacturing **1306** and system integration **1308** in FIG. **13**. One or more apparatus embodiments, method embodiments, or a combination thereof may be utilized while aircraft **1400** is in service **1312**, during maintenance and service **1314**, inclusive of inspection, in FIG. **13**, or both. The use of a number of the different illustrative embodiments may substantially expedite the assembly of aircraft **1400**, reduce the cost of aircraft **1400**, or both expedite the assembly of aircraft **1400** and reduce the cost of aircraft **1400**.

(86) The description of the different illustrative embodiments has been presented for purposes of illustration and description and is not intended to be exhaustive or limited to the embodiments in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other desirable embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A nose gear spin down apparatus, comprising: a spin down device rotatable about a central longitudinal axis, wherein the spin down device rotates about the central longitudinal axis at a first

speed after engagement with a spinning nose gear wheel of an aircraft that spins at a second speed; the spin down device comprising: a tubular casing comprising internal threads; and a threaded shaft positioned within the tubular casing and comprising external threads engaged with the internal threads of the tubular casing; and a resistance device distanced from the spin down device by a space; wherein the resistance device imparts a resistance to the spin down device when movement of the threaded shaft relative to the tubular casing along the central longitudinal axis or movement of the tubular casing relative to the threaded shaft along the central longitudinal axis causes the threaded shaft or the tubular casing to move through the space and into contact with the resistance device.

2. The nose gear spin down apparatus of claim 1, wherein the resistance device provides no resistance to the spin down device until the spin down device moves through the space and contacts the resistance device.

3. The nose gear spin down apparatus of claim 1, wherein the resistance device is a spring.

4. The nose gear spin down apparatus of claim 1, wherein the resistance device is a series of axially aligned springs, wherein each spring of the series of axially aligned springs has a different stiffness.

5. The nose gear spin down apparatus of claim 1, wherein the resistance device is a viscous damper.

6. The nose gear spin down apparatus of claim 1, wherein the resistance of the resistance device applied to the spin down device progressively builds after the spin down device contacts the resistance device.

7. The nose gear spin down apparatus of claim 1, further comprising a locking mechanism engaged with the resistance device to prevent release of an energy captured by the resistance device during engagement of the spin down device with the spinning nose gear wheel of the aircraft.

8. The nose gear spin down apparatus of claim 1, wherein the tubular casing comprises a machined or adhesive outer surface for engagement with the spinning nose gear wheel of the aircraft.

9. The nose gear spin down apparatus of claim 1, further comprising a wheel mounted to the tubular casing, the wheel for engagement with the spinning nose gear wheel of the aircraft.

10. The nose gear spin down apparatus of claim 1, wherein the tubular casing rotates relative to the threaded shaft.

11. The nose gear spin down apparatus of claim 1, wherein the threaded shaft rotates relative to the tubular casing.

12. The nose gear spin down apparatus of claim 1, further comprising: a wheel mounted to an axle; a first gear set mounted on the axle, the first gear set comprising a forward gear fixed to a reverse gear, wherein the first gear set is slidable longitudinally along the axle between a forward position and a reverse position; and a second gear set mounted on the threaded shaft; wherein when the first gear set is in the forward position, the forward gear of the first gear set engages the second gear set and wherein when the first gear set is in the reverse position, the reverse gear of the first gear set engages an idler gear and the idler gear engages the second gear set.

13. A nose gear spin down apparatus, comprising: a tubular casing comprising internal threads; a threaded shaft positioned within the tubular casing and comprising external threads engaged with the internal threads of the tubular casing; and a resistance device axially aligned with the threaded shaft, wherein the tubular casing and the threaded shaft and the resistance device share a central longitudinal axis; wherein movement of the threaded shaft relative to the tubular casing along the central longitudinal axis or movement of the tubular casing relative to the threaded shaft along the central longitudinal axis engages the resistance device after the threaded shaft or the tubular casing moves through a space; wherein the tubular casing rotates relative to the threaded shaft or the threaded shaft rotates relative to the tubular casing.

14. The nose gear spin down apparatus of claim 13, further comprising: a wheel mounted to an axle; a first gear set mounted on the axle, the first gear set comprising a forward gear fixed to a reverse gear, wherein the first gear set is slidable longitudinally along the axle between a forward position and a reverse position; and a second gear set mounted on the threaded shaft; wherein when

the first gear set is in the forward position, the forward gear of the first gear set engages the second gear set and wherein when the first gear set is in the reverse position, the reverse gear of the first gear set engages an idler gear and the idler gear engages the second gear set.

15. The nose gear spin down apparatus of claim 13, wherein the resistance device provides no resistance to the threaded shaft or the tubular casing until the threaded shaft or the tubular casing moves through the space and contacts the resistance device.

16. The nose gear spin down apparatus of claim 13, wherein the resistance device is a spring.

17. The nose gear spin down apparatus of claim 13, wherein a resistance of the resistance device applied to the threaded shaft or the tubular casing progressively builds after the threaded shaft or the tubular casing contacts the resistance device.

18. A method for slowing rotation of a nose gear wheel of an aircraft after takeoff, the method comprising: retracting a spinning nose gear wheel of the aircraft to a stowed position in a nose gear bay of the aircraft after takeoff of the aircraft; engaging the spinning nose gear wheel of the aircraft with a nose gear spin down apparatus mounted in the nose gear bay of the aircraft without imparting a resistance to the spinning nose gear wheel of the aircraft; and imparting the resistance to the spinning nose gear wheel of the aircraft with a resistance device of the nose gear spin down apparatus after an element of the nose gear spin down apparatus moves through a space between the element of the nose gear spin down apparatus and the resistance device and into contact with the resistance device.

19. The method of claim 18, further comprising: storing energy from the spinning nose gear wheel of the aircraft as the nose gear spin down apparatus imparts the resistance to the spinning nose gear wheel of the aircraft and slows rotation of a forward direction of the spinning nose gear wheel of the aircraft after takeoff; and prior to deploying the spinning nose gear wheel of the aircraft from the stowed position, releasing the energy to the spinning nose gear wheel of the aircraft to rotate the spinning nose gear wheel of the aircraft in the forward direction.

20. The method of claim 18, wherein the element of the nose gear spin down apparatus is a threaded shaft, the nose gear spin down apparatus further comprises: a tubular casing comprising internal threads, wherein engagement of the spinning nose gear wheel of the aircraft with the tubular casing causes rotation of the tubular casing about a central longitudinal axis of the tubular casing; the threaded shaft is positioned within the tubular casing and is engaged with the internal threads of the tubular casing, wherein rotation of the tubular casing causes lateral movement of an end of the threaded shaft through the space; and the resistance device is axially aligned with the threaded shaft, wherein rotation of the tubular casing about the central longitudinal axis urges the end of the threaded shaft to move laterally with respect to the tubular casing along the central longitudinal axis through the space to contact the resistance device, wherein the resistance device provides the resistance to the lateral movement of the threaded shaft which translates the resistance to the rotation of the tubular casing.
